



CHAT GPT

Rewriting the Rules of OpenAI's Novel Revenue Model for the GPT Store and Strategic Chip Alliances

~ A Case Study by team “Case Ace”

TABLE OF CONTENTS

- 1. Overview
- 2. Industry Analysis
- 3. GPT Store
- 4. User Journey
- 5. Revenue Model
- 6. GTM Strategy
- 7. Revenue Model Analysis
- 8. Q2 . Overview
- 9. Current Scenario
- 10. Comparison
- 11. Final Conclusion
- 12. Executive Summary



OpenAI scraps transaction fee model for GPT Store, seeks alternative Revenue streams

10 Nov 2023, Alex Jones, The Guardian

Will Low-quality bots flood the Market? OpenAI's GPT Store monetization strategy questioned

22 Dec 2023, Ms. Sophia Lee, AI Journalist, BBC)

OpenAI's GPT Store: A Boon for Creators or a Cash Grab? The Debate Begins"

13 Oct 2023, John Smith, The New York Times

PROBLEM STATEMENT

- Detailed Revenue model analysis:** Explore specific revenue options, pricing strategies, and cost projections
- Financial Feasibility:** Analyze the proposed model's potential revenue streams, costs, and profitability in the short and long term
- Go-to-Market strategy:** Develop competitive strategy to differentiate GPT Store and attract users in a crowded AI market
- Quality Control:** Implement strategies to prevent oversaturation and ensure the development of high-quality, specialized, and sustainable GPTs that cater to diverse user needs

79% People had some exposure to generative AI and

22% Users use generative AI regularly in their own work

92% Fortune 500 company build on Chat GPT products

1.4B OpenAI website visits per month

100M Weekly active users on Chat GPT

10M+ Daily user queries

OVERVIEW

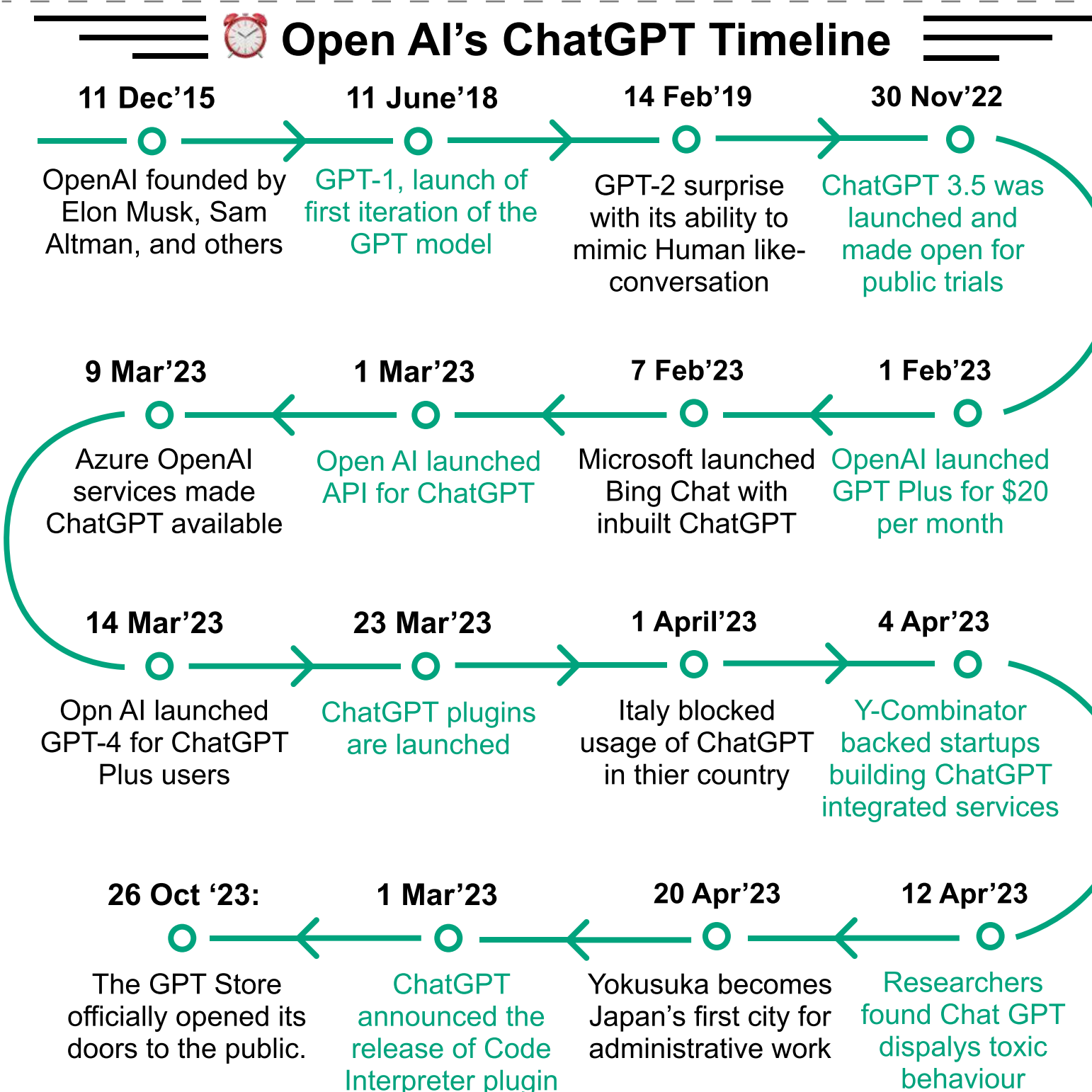
ChatGPT : Learning the Ropes before diving into the GPT Store

What is Chat GPT?

Chat GPT (Chat Generative Pre-Trained Transformer) is a large language model chatbot developed by **OpenAI**, launched in November 2022 that is trained on a massive **dataset of text and code**, allowing it to **mimic human** conversation and generate realistic and creative text formats.

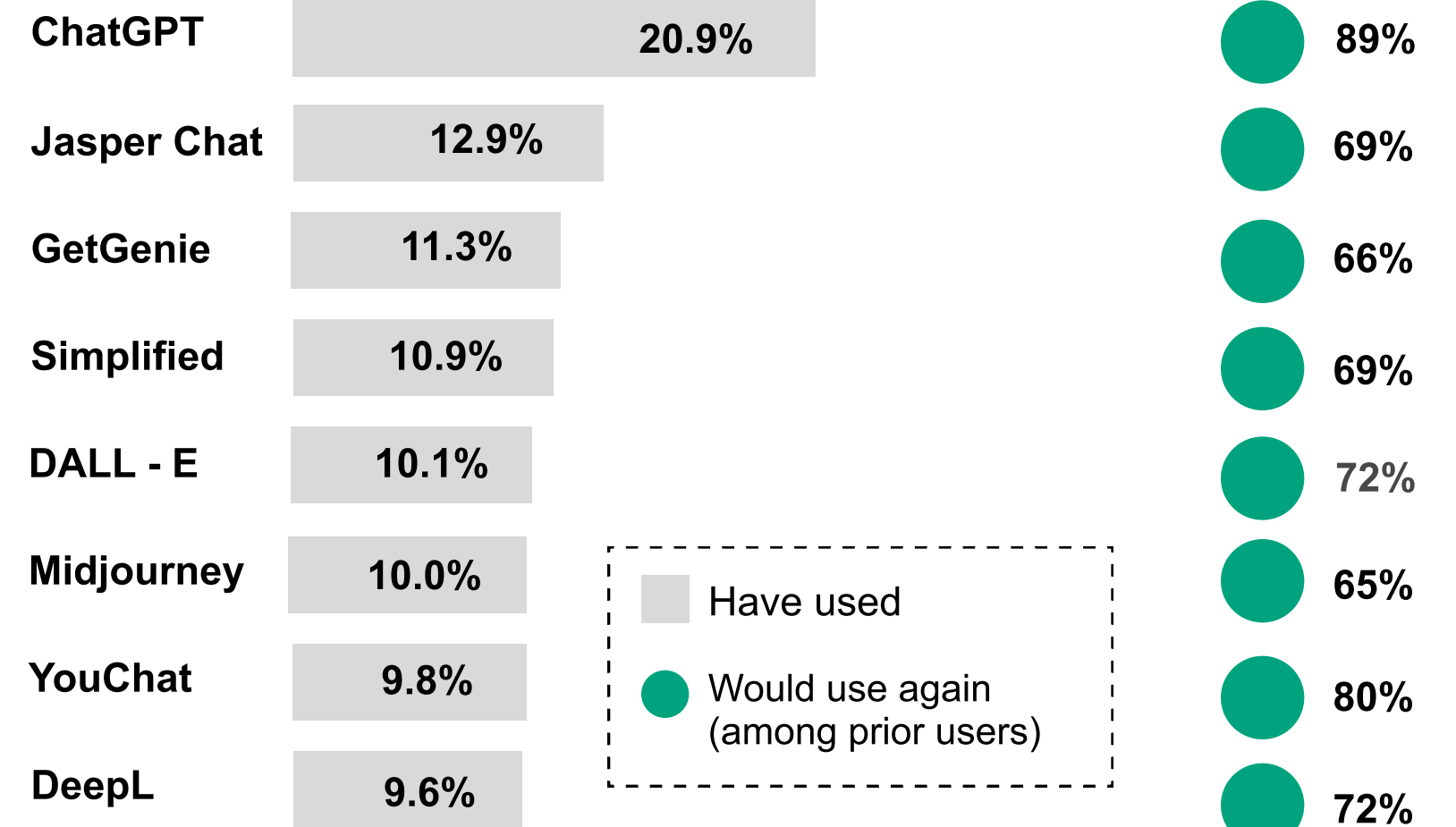
Versions of Chat GPT

- ChatGPT** : Initial versatile conversational AI release
- GPT-4 Turbo**: Economical, efficient version for diverse AI tasks
- GPTs** : Empowers users to train models for dynamic tasks
- GPT Store & GPT Builder**: Collaborative AI marketplace with easy model creation
- Assistants API**: Key tool for developers, enhancing AI applications



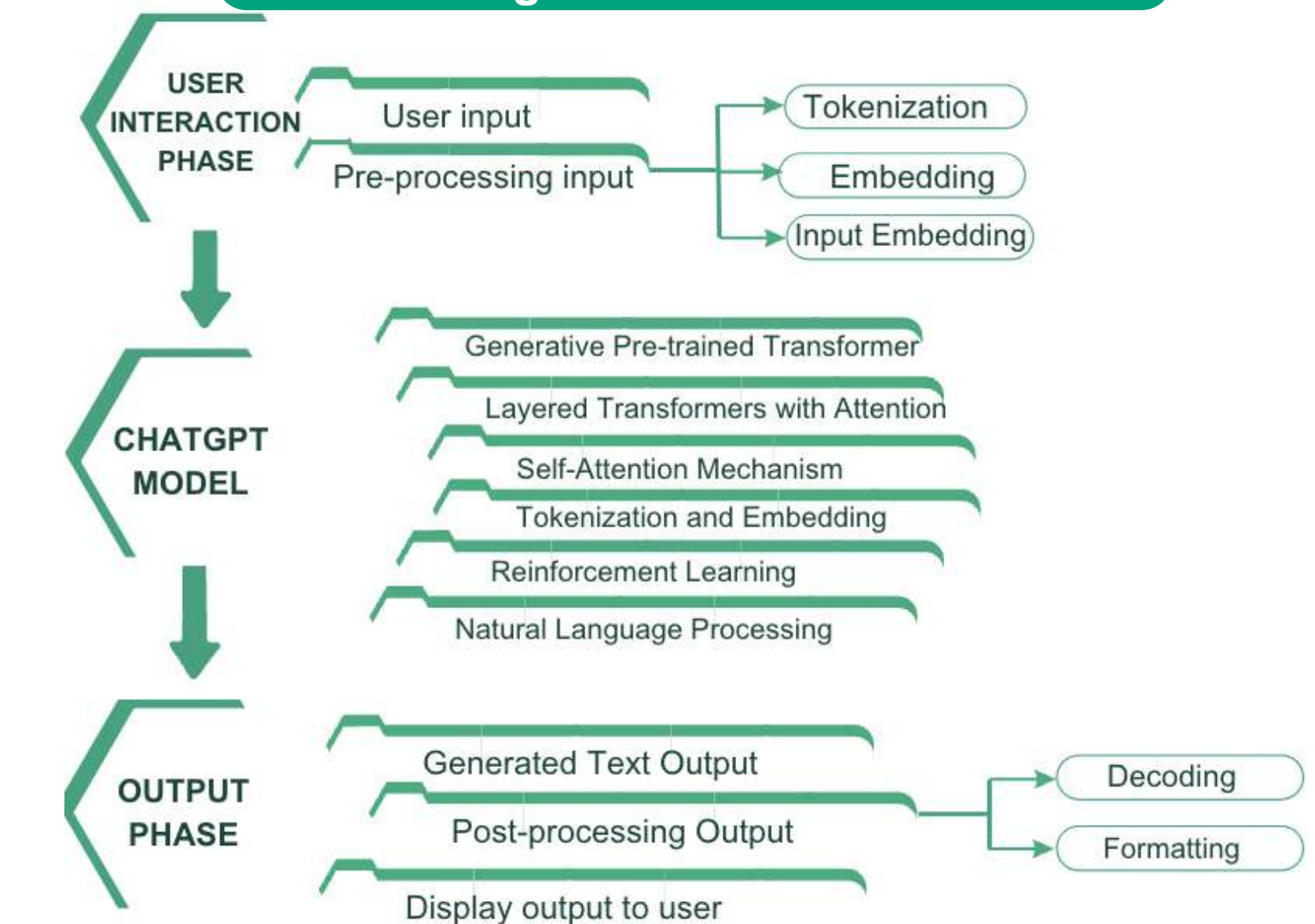
ChatGPT- The Most tried AI tool and users stick to it

Share of U.S. respondents who have used the following AI tools in the past 12 months



1237 U.S respondents (16-64 y/o) surveyed Mar-Aor 2023 (Source: Statista)

Working Flowchart of Chat GPT



The AI Industry confirms the explosive growth of Generative AI (gen AI) tools

AI Industry

Current Industry size in 2023
\$150.2 B

Projected Industry size by 2032 (CAGR - 19.4%)
\$2,575.2 B

Generative AI Industry

Current Industry size in 2023
\$10.14 B

Projected AI Industry size by 2032 (CAGR - 19.4%)
\$191.8 B

The foundation of **OpenAI** in 2015

MISSION

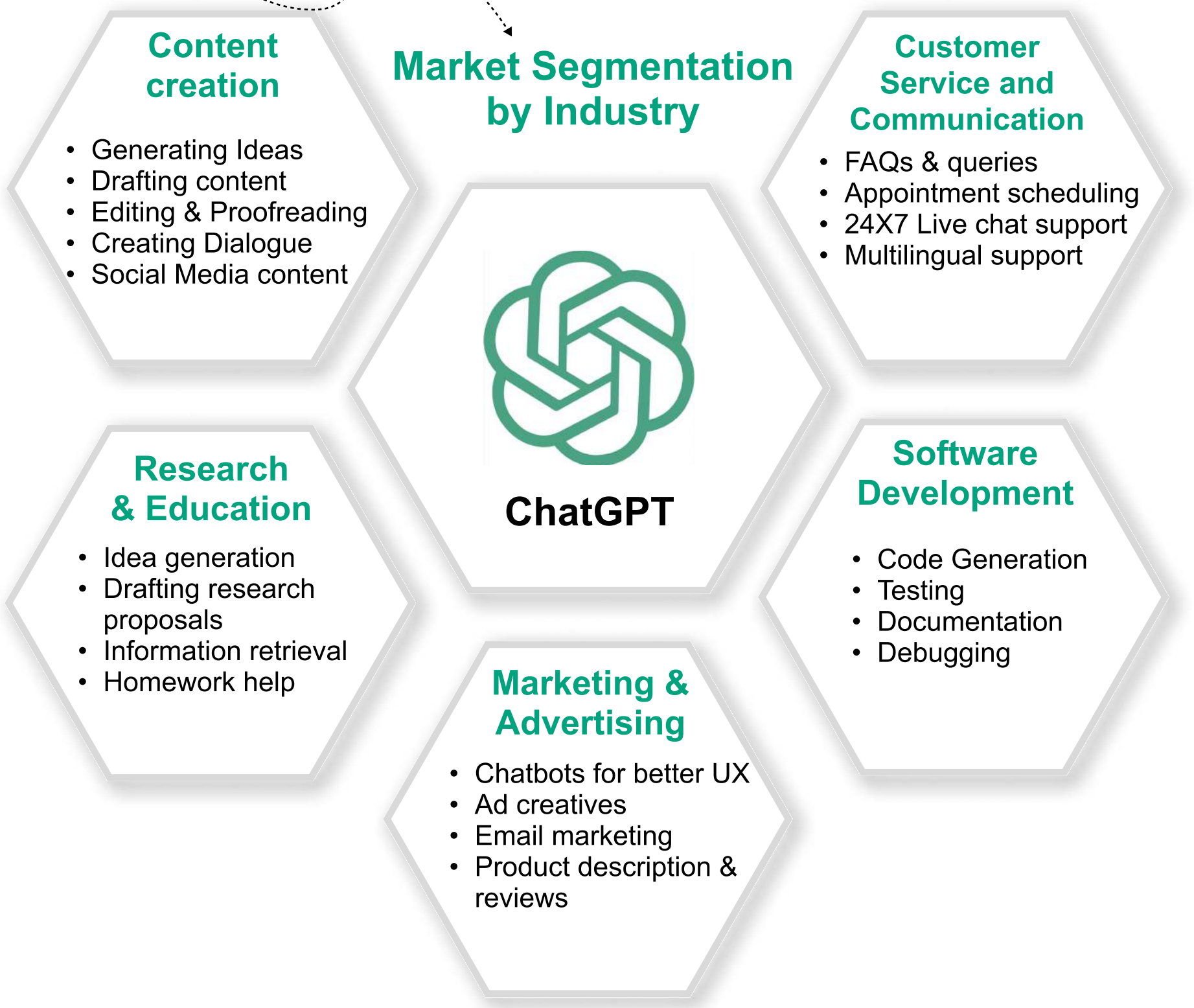
Ensure that Artificial General Intelligence (AGI) benefits for all with a focus on safety of humanity



VISION

To achieve a future where AGI is used for good, contributing to the well-being and progress of humanity.

Developed & launched ChatGPT to enter Generative AI Industry



Generative AI Industry: Market Segmentation - Technology type

	Large Language Models (LLMs)	Generative Adversarial Networks (GANs)	Autoregressive Models
FOCUS	Text generation, translation, code generation, question answering, writing different kinds of creative content	Image and video generation, editing, manipulation, style transfer, data augmentation	Text and music generation, code generation, natural language inference
APPLICATIONS	Content creation, customer service, education, research, software development	Media & entertainment, fashion, healthcare, manufacturing, design	Music composition, code generation and completion, creative writing
MARKET SIZE	Current Market size in 2022 \$85 M Projected Market size by 2032 \$1.3 T	Current Market size in 2023 \$423 M Projected Market size by 2028 \$1.4 B	Market size in 2020 \$1.73 B Projected Market size by 2026 \$3.29 B
GROWTH	Text-focused tasks, personalized content, multi-modal generation	Photorealistic image and video generation, medical imaging analysis, product design	Music creation software, software development tools, creative content platforms
KEY PLAYERS	OpenAI (ChatGPT) , Google (LaMDA), Microsoft (Bard), Baidu (ERNIE), DeepMind (PaLM)	NVIDIA StyleGAN2, DeepMind BigGAN, Adobe Project Faceless, RunwayML	Jukebox by OpenAI, MuseNet by Google, GPT-2 by OpenAI (predecessor to GPT-3)

INDIVIDUALS

Writers, artists, programmers, researchers, anyone seeking assistance with creative or analytical tasks.

WRITERS**PROGRAMMERS****RESEARCHERS**

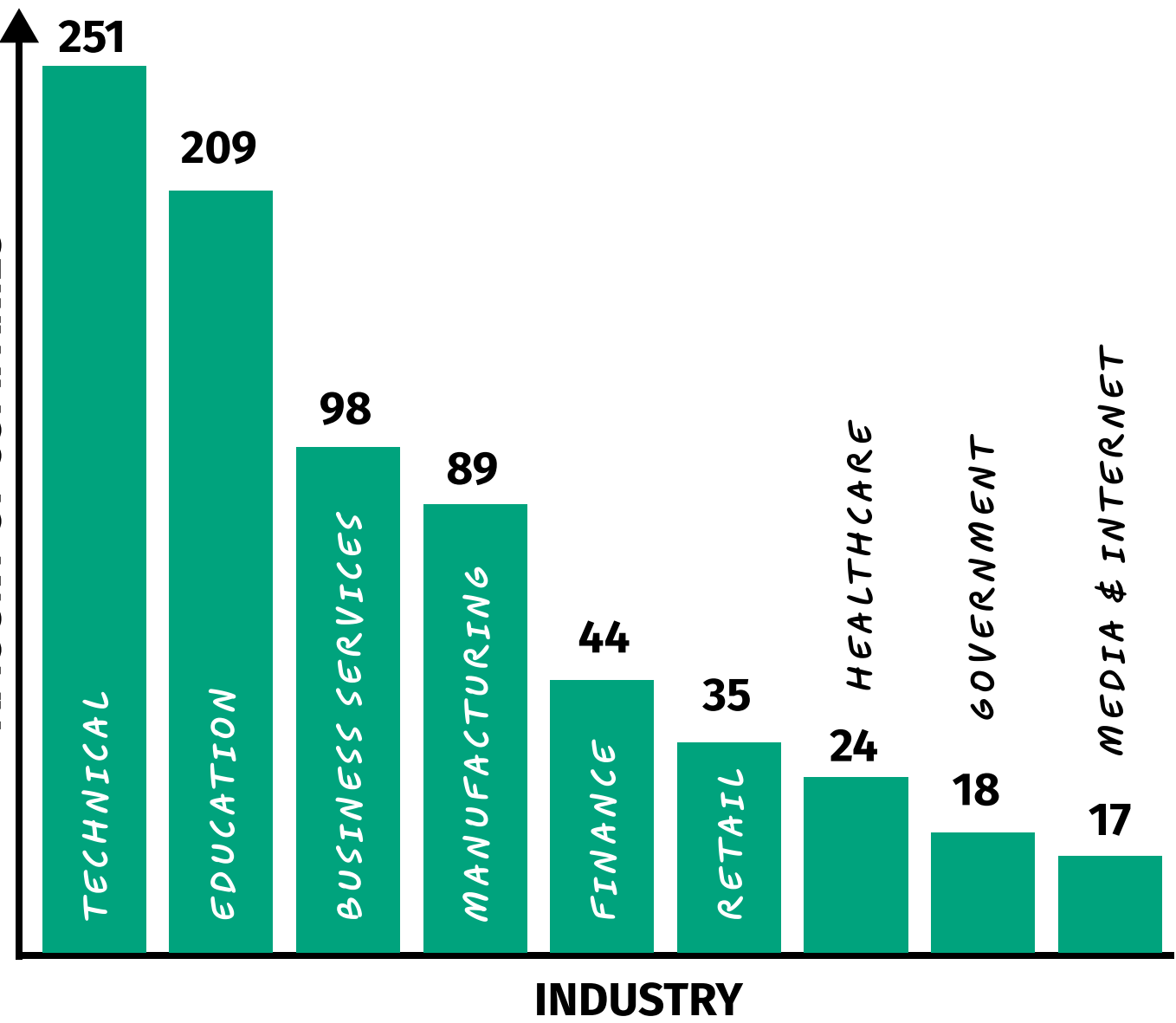
BUSINESS

Marketing agencies, media companies, software developers, organizations looking for content creation, chatbot development, or code assistance.

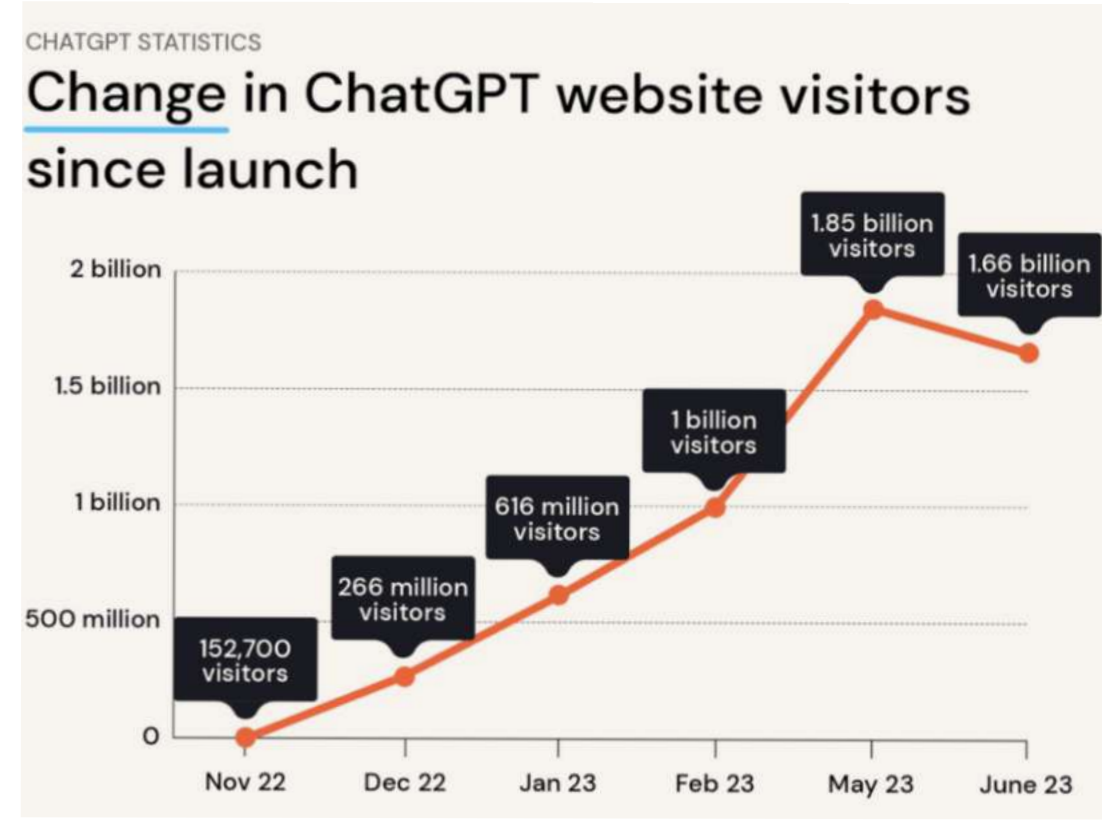
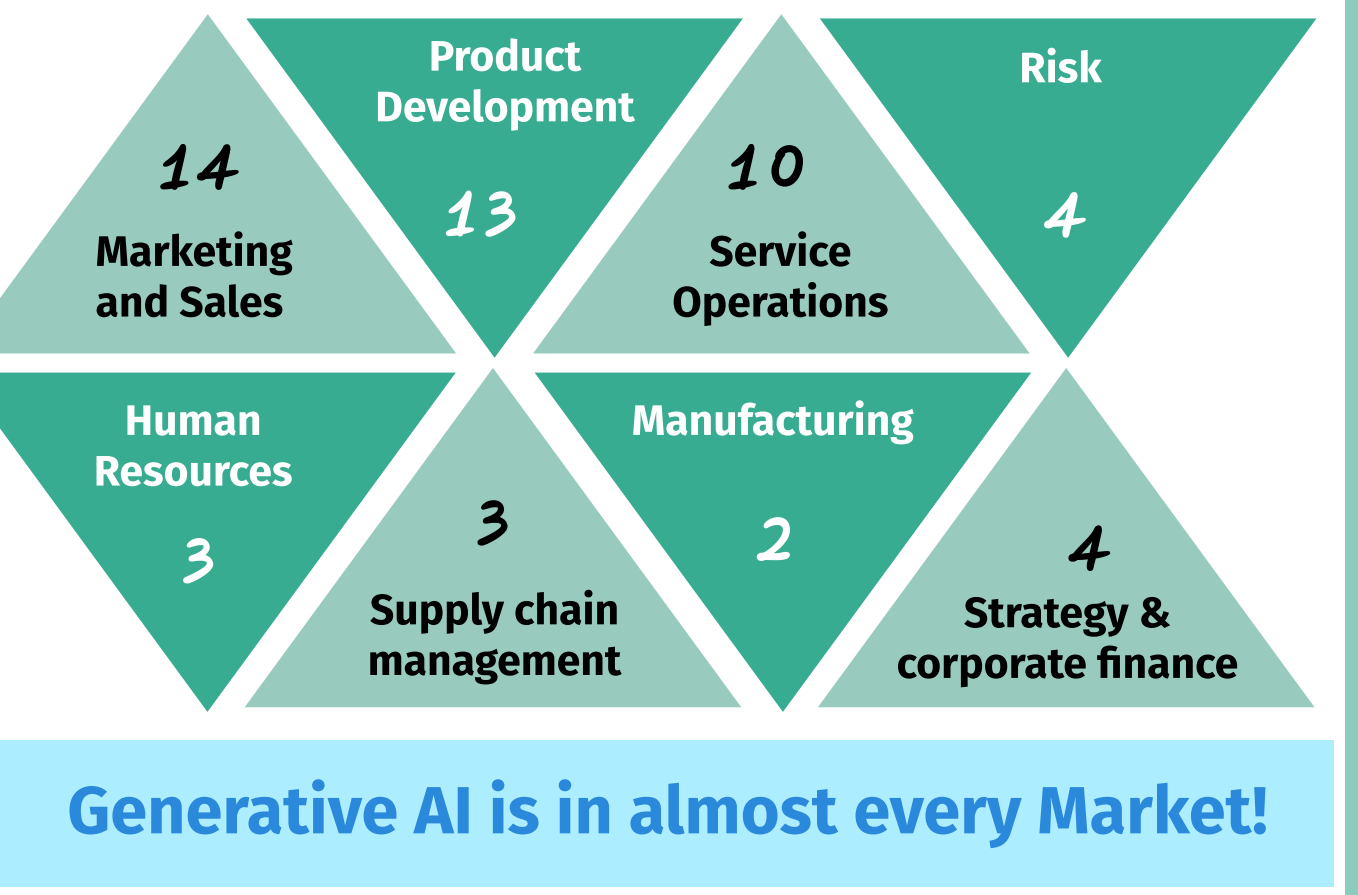
MARKETING AGENCY**CUSTOMER SERVICE****CONTENT CREATORS**

180.5 M USERS

Amount of companies using ChatGPT in thier business function In 2023, by INDUSTRY



% Share of respondents reporting that their organization is regularly using generative AI in given function



Possible reasons for decrease in user base after May'23

Technical Issues: In July 2023, ChatGPT experienced technical issues causing downtime and access limitations. This temporary disruption could have further contributed to the decline in user engagement during that period.

Newness Wearing Off: ChatGPT launched in November 2022 and saw an initial surge in popularity due to its novel abilities and public interest in large language models. However, by May 2023, the novelty could have worn off for some users, leading them to explore other tools or lose interest in general.

Most regularly reported generative AI use cases within function, % of respondents

MARKETING & SALES

- Crafting first drafts of text documents (9%)
- Personalized marketing (8%)
- Summarizing text documents (9%)

PRODUCT DEVELOPEMENT

- Identifying trends in user needs (7%)
- Drafting technical documents (5%)
- Creating new product designs (4%)

SERVICE OPERATIONS

- Use of Chatbots (eg,customer service) (6%)
- Forecasting service trends or anomalies (5%)
- Creating first drafts of documents (5%)

COMPETITIVE ANALYSIS

Currently, no platform directly offers a marketplace for custom LLM models similar to Open AI's **GPT Store**. The GPT Store might be the first, but it won't be the last. Let's meet the gladiators entering the LLM arena!

Competitor	Core Offerings	Competitive Advantages	Potential Weaknesses	Impact on GPT Store
 Google AI Vertex	- Access to pre-trained LLMs (e.g., LaMDA) - Custom model training capabilities - Integration with Google Cloud Platform	- Established cloud infrastructure - Diverse pre-trained models - Comprehensive development tools	- Focus on enterprise users - High pricing for custom models - LeAss user-friendly interface	Potentially capture high-budget users, but user experience and ethical considerations might lag behind GPT Store
 Microsoft Azure Cognitive Services	- Pre-trained LLMs for text generation, translation, and other tasks - Custom model training through Azure Machine Learning	- Broad range of AI services - Integration with Microsoft ecosystem	- Limited focus on custom LLMs - Complex technical requirements for model development	Offer alternative for specific Azure users, but lack of curated marketplace might limit appeal
 Amazon SageMaker	Cloud-based platform for building, training, and deploying custom AI models, including LLMs	- Scalable infrastructure - Cost-effective options for development	- Less user-friendly interface for non-technical users - Limited curated LLM offerings	Potentially attract technical users interested in building their own models, but lack of curated platform might discourage broader adoption
 Emerging LLM Startups (Cohere, Anthropic)	- Custom LLM development and deployment for various industries - Open-source research on LLMs with focus on safety and responsible AI	- Expertise in LLM development - Strong partnerships with businesses (Cohere) - Advanced research capabilities (Anthropic)	- High cost of custom models (Cohere) - Limited commercial offerings (Anthropic) - Less user-friendly tools and resources (Anthropic)	- Potential threat for high-value projects (Cohere) - Could influence ethical standards and research (Anthropic) - Not direct competitors in the user-facing market



Imagine a marketplace where anyone can access and unlock the power of custom **Large Language Models (LLMs)**

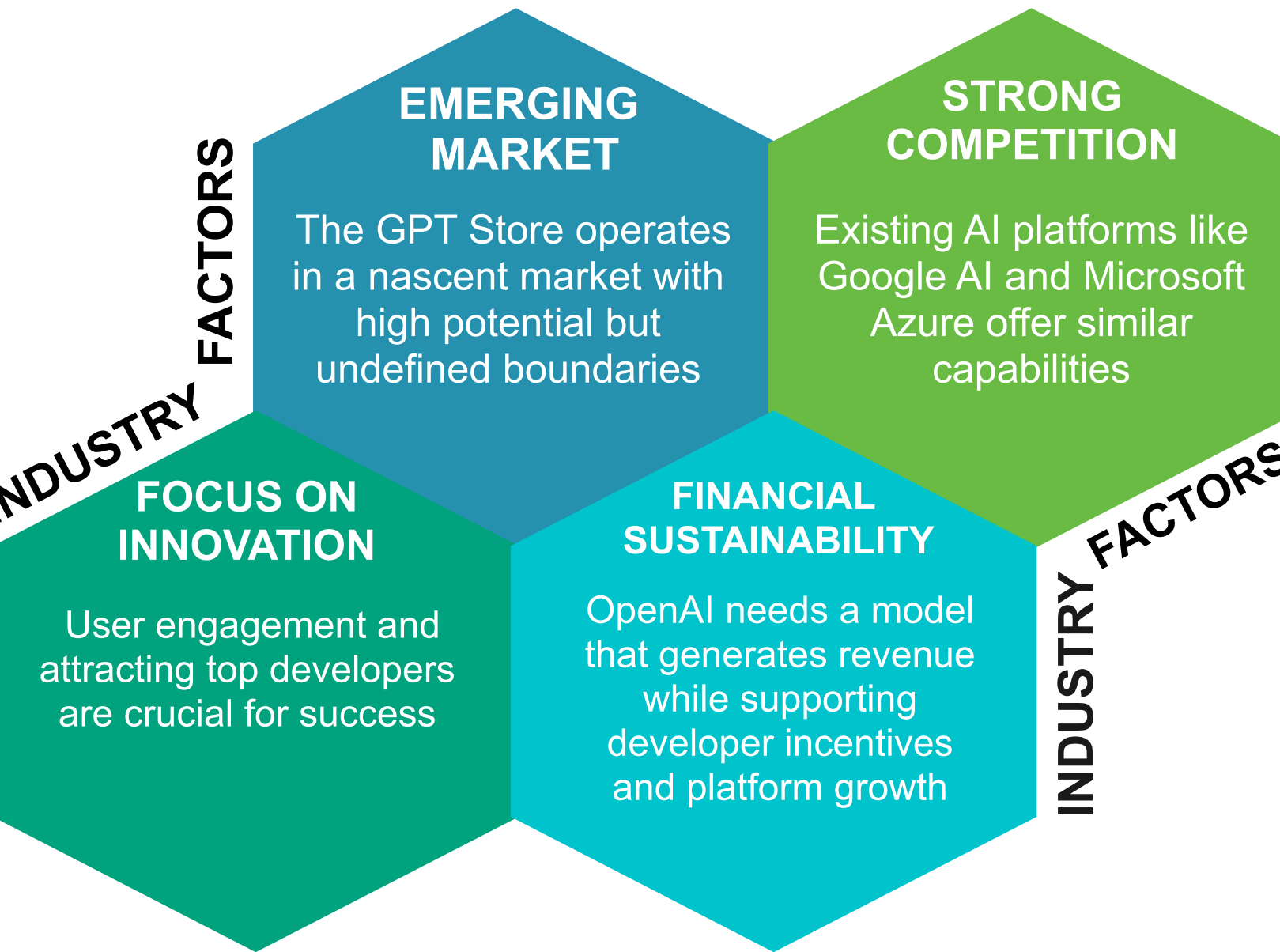
WHAT IS GPT STORE?
The GPT Store is a platform for **Developers** to share custom models for diverse purposes. **Users** can browse, explore, and test models before integrating them into projects.

VALUE PROPOSITION

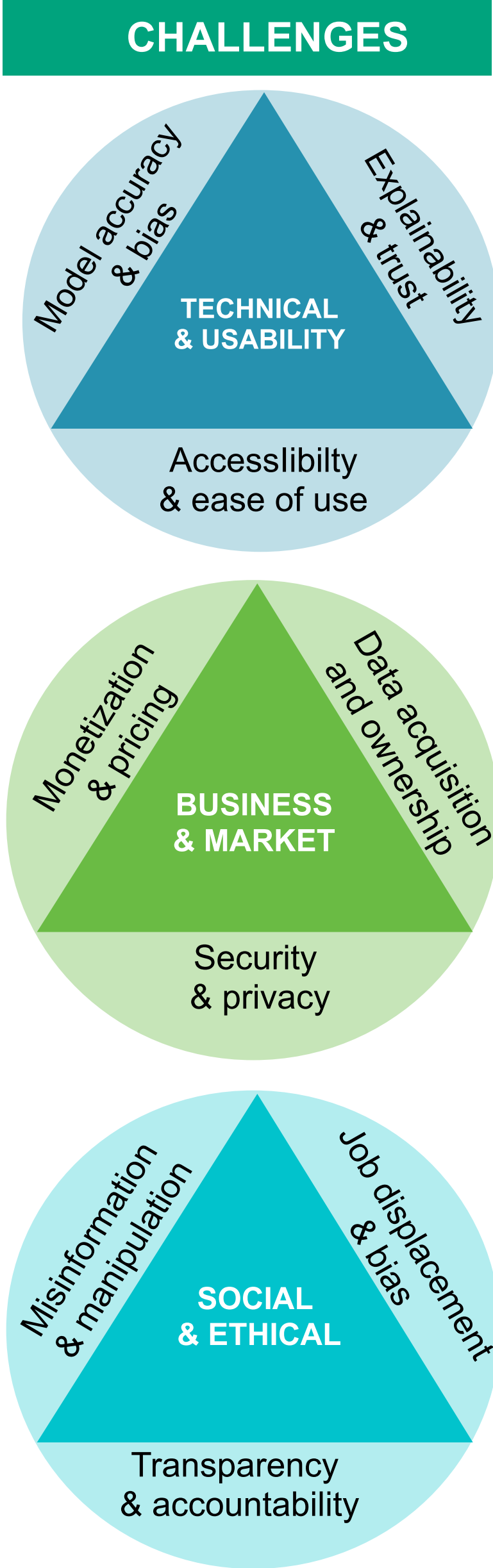
Unleash Creativity and Productivity

Empower Everyone with AI Expertise

Bridge Gap b/w Data and Insight



CURRENT STATUS
Under development, undergoing closed beta testing with launch planned for 2024



USER PERSONA'S



Satish & Shashi Godbole Product based Startup founders • Price-Conscious • Busy

CORE NEEDS & GOALS

- Run Startup strategically & attend Meetings
- Market research and validation

PAIN POINTS

- Need someone to write Product descriptions
- No one there for 24x7 Customer Service



Vidya Venkatesan College Graduate • Web Developer • Freelancer

CORE NEEDS & GOALS

- Meet Deadlines & Stay creative
- Generate fresh Ideas & ensure uniqueness

PAIN POINTS

- Juggle b/w multiple projects
- Do repetitive task & Time Pressure



Ishaan Awasthi 12 Year Old • School Student • Introvert

CORE NEEDS & GOALS

- Wishes to impress his teacher by speaking fluent english
- Wishes to complete his homework asap to be able to play

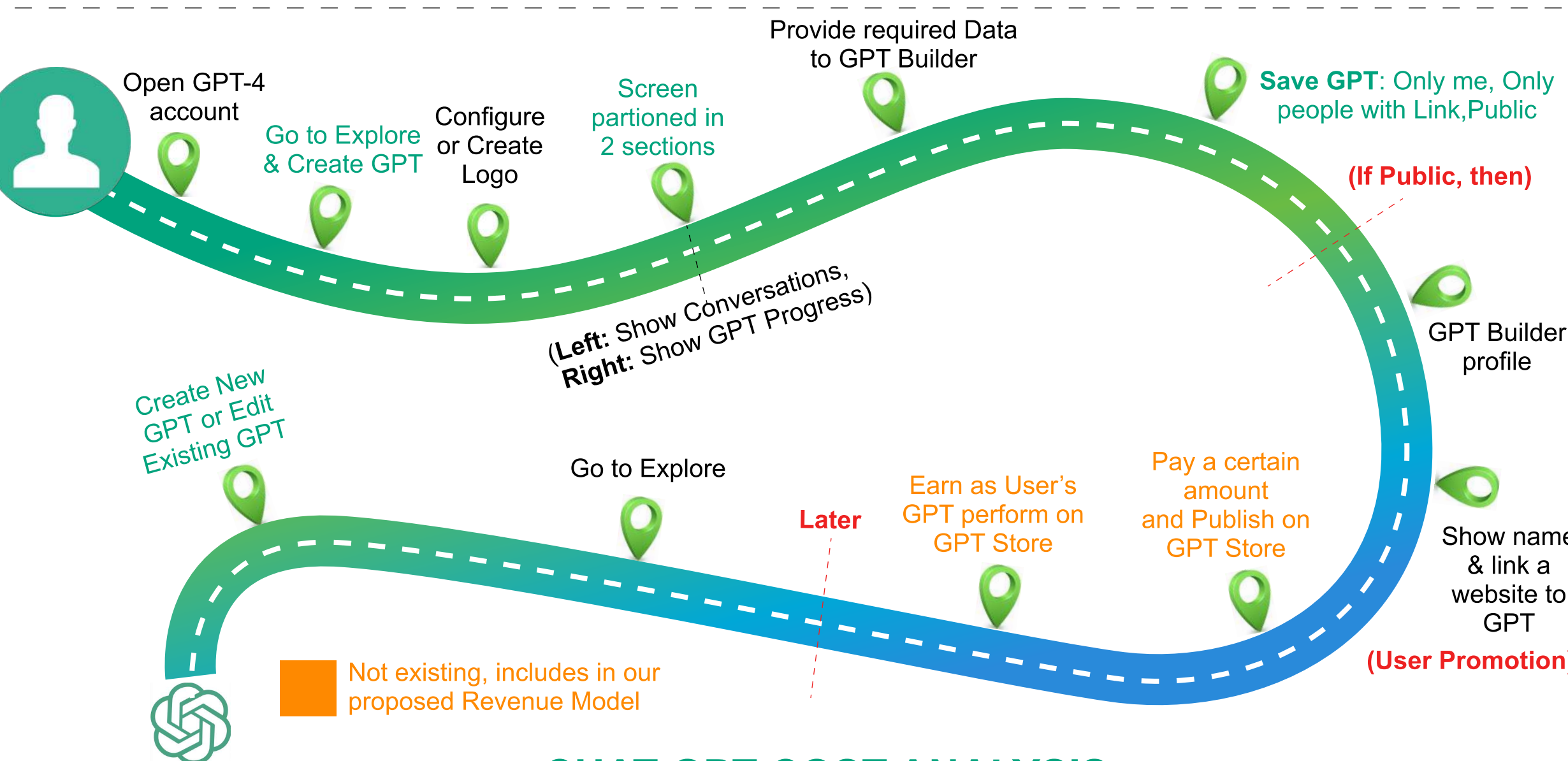
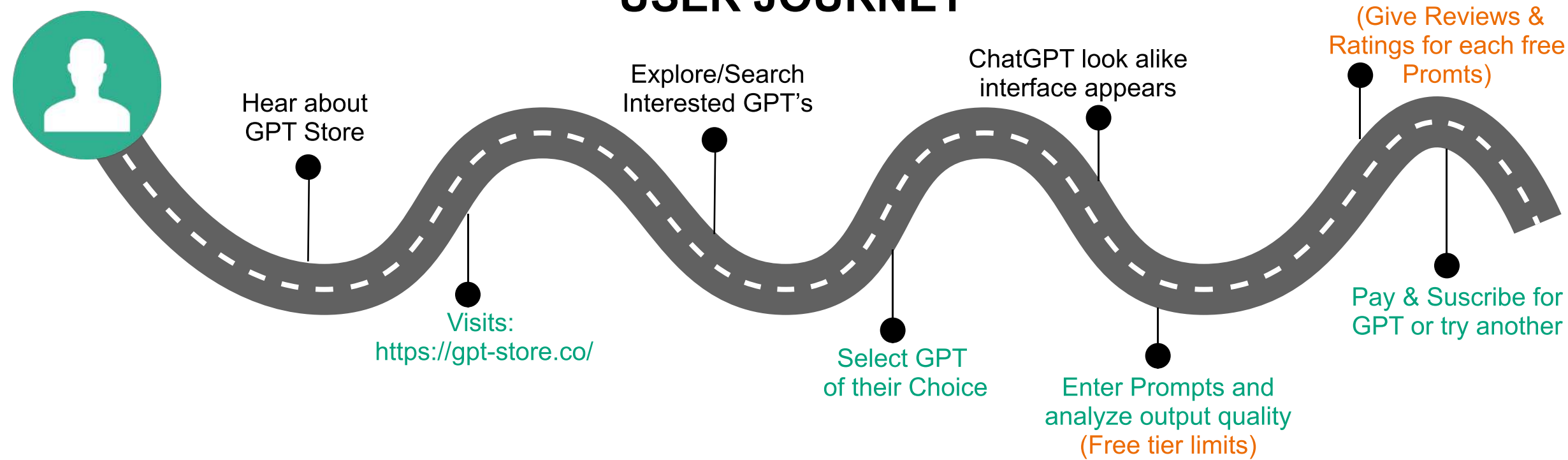
PAIN POINTS

- Unable to communicate his difficulties in concepts
- Dislikes being told to not be on his iPad

A vast variety of Users can be included in Generative AI's market

GPT Store

USER JOURNEY



CHAT GPT COST ANALYSIS

Cost of each word generated on ChatGPT = **\$0.0003**

Average word generation per response = **150**

No. of Daily response : **More than 10,000,000**

\$0.0003 X 133 = \$0.04 per response

Daily running cost of ChatGPT = **\$400,000 +**

Monthly running cost of Open AI = **\$12M**

Cost of each Image generation (DALL-E 2) = **\$0.02**

Cost of each **HD** Image generation (DALL-E 3) = **\$0.08**

Huge Operating Cost, then how Chat GPT gives free tier access to everyone?

Let's Explore



USER JOURNEY & REVENUE MODEL

GPT PRICING

Free

\$0 per person/month

[Try it now ↗](#)

For people just getting started with ChatGPT

- ✓ Unlimited messages, interactions and history
- ✓ Access to GPT 3.5 model
- ✓ Access on Web, iOS, and Android

Plus

\$20 per person/month

[Upgrade now ↗](#)

Everything in Free, and:

- ✓ Access to GPT-4
- ✓ Browse, create, and use GPTs
- ✓ Access to additional tools like DALL-E, Browsing, Advanced Data Analysis and more

REVENUE STREAMS OF CHAT GPT

While ChatGPT offers a free tier, it's essential to remember that it's not entirely free to operate. Here's how OpenAI covers costs and generates revenue while providing free access:

PAID TIERS

Businesses & developers pay substantial fees for premium access with higher query limits, longer responses, fine-tuning, and more. This revenue subsidises the costs of free users.

RESEARCH PARTNERSHIPS

OpenAI partners with institutions and companies on paid research projects, utilising ChatGPT and generating income.

LICENSING

OpenAI licence its technology, including parts of ChatGPT, to other companies for integration into their products and services. This expands their reach and influence

GRANTS & INVESTMENT

OpenAI receives funding from sources like Microsoft, reducing its reliance on direct user payments

How Open AI Manage Free Tier Costs?

Quotas and limitations

The free tier has usage restrictions (e.g., query limits, response length) to manage costs and prevent overload

Potential future monetization

OpenAI may explore future monetization models for free tier (eg. advertising or data collection)

REVENUE MODEL NEED GPT STORE

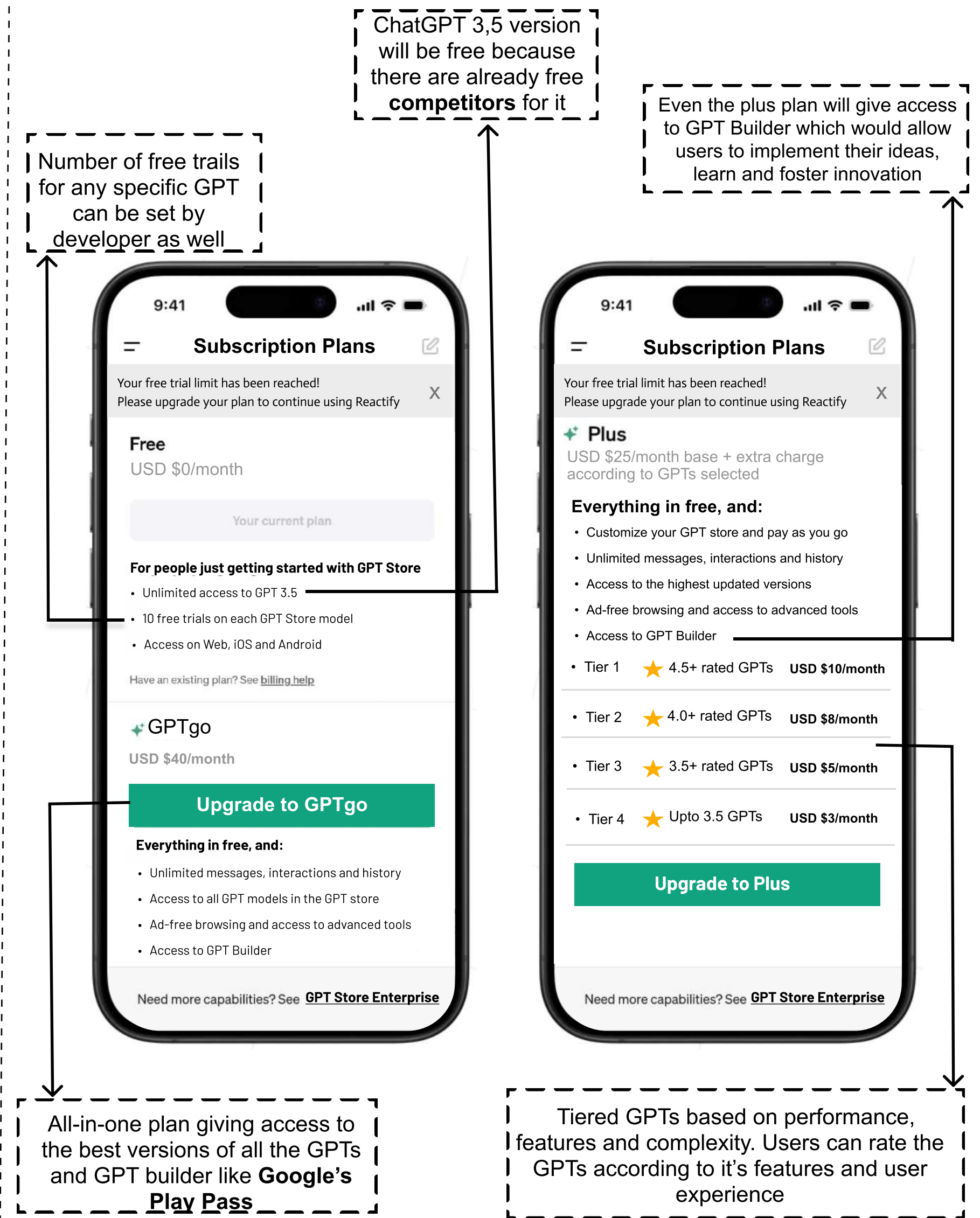
- GPT Plus Plan at \$20 grants access to all GPTs.
- OpenAI is close to break-even point despite launching ChatGPT 3, ChatGPT 3.5 (freemium), and ChatGPT 4
- Aim to incentivize developers to monetize GPTs, fostering innovation and expanding AI.
- Proposing a new revenue model to shift from non-profit to profitable expansion.
- Transition benefits both developers and OpenAI, contributing to AI community growth.
- To answer the accessibility to the powerful GPT Builder be restricted to paid users, or should some level of functionality be available to everyone?

1 Subscription from User

Freemium Features



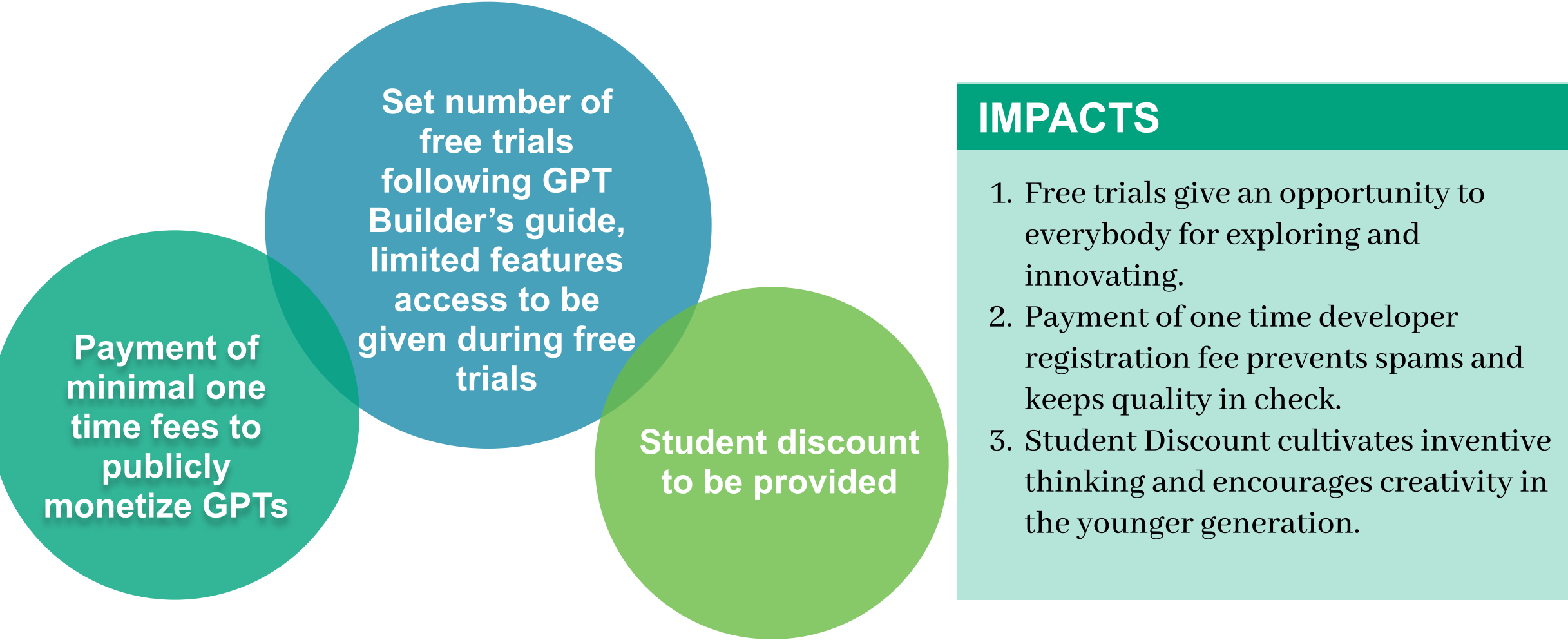
Pricing Model



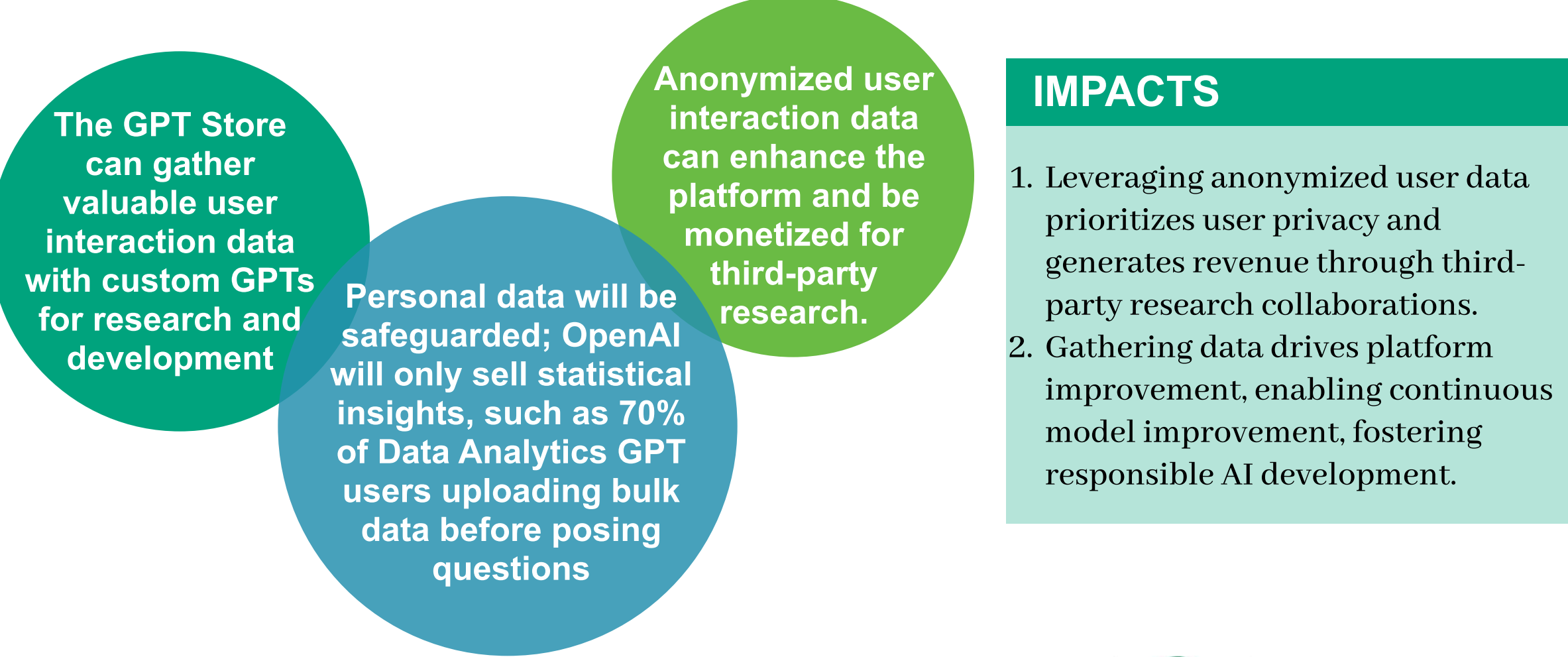
REVENUE MODEL



2 ONE TIME PAYMENT FROM DEVELOPERS



3 DATA ANALYTICS AND IT'S MONETIZATION



4 ADVERTISEMENTS

According to available information, approximately 99.9% of the user base utilizes the free model, while **only 0.1%** are paid customers, so keeping this in mind advertisement can be a good option to generate, at a cost of **compromising a little with User Experience**.

NO ADVERTISEMENT	ADVERTISEMENT
• Enhanced User Experience: Users experience Clean UI/UX and are more likely to trust content if they believe it is free from commercial influence • For applications emphasizing information, education, or non-commercial entertainment, excluding advertisements may be more appropriate.	• The inclusion of advertisements can serve as a revenue stream, helping offset development and maintenance costs. • Advertisements could be implemented only if Open AI will be at a verge of Bankruptcy in future, till then they should focus on Clean UI/UX

Integrating Ads directly within the model's outputs at strategic points, after text generation.
Eg. If someone ask for best the Website Wordpress Development site, ChatGPT can show the advertised sites as a priority but with a “Promoted” Tag, a “Response without Promotions” button will pop up at such type of responses, giving user flexibility.

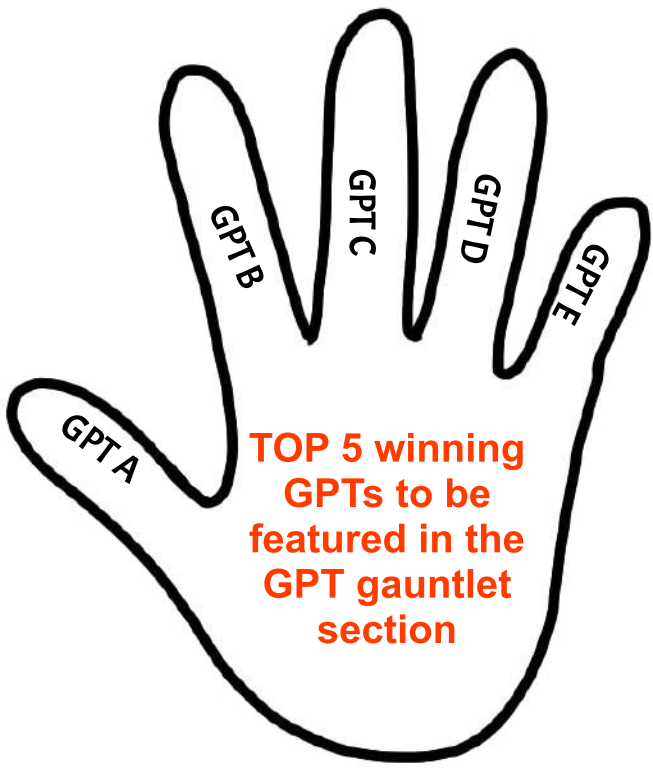
Relevant Advertisement at page corners in GPT Store “Categories” section
Include **relevant categorized** Advertisements to minimize interference in quality user experience

5 GPT GAUNTLET

To foster innovation in AI development...

A monthly global challenge where developers get the opportunity to showcase their GPT building capabilities, competing for alluring rewards and recognition.

Setting a nominal registration fee of \$2 or \$3 (**low fees**) can serve as a motivator for individuals to engage in creative and innovative endeavors. The collected funds can be used to finance prizes, ensuring a small profit margin for the company as well.



GO-TO-MARKET STRATEGY

USER SEGMENTATION

Freemium Users:

- The largest segment (around 200M)
- Primarily casual users, content consumers, and early experimenters with GPTs.

Pay-as-You-Go Users:

- Value-conscious users requiring specific GPT functionalities for occasional tasks.

GPTgo Subscribers:

- Power users and professionals seeking unlimited access and premium GPT experiences.

Developers: GPT creators and entrepreneurs generating revenue from selling custom GPTs in the store.

Developers:

- Will get paid on every purchase of their GPT on commission basis
- Competitions offering incentives
- Provide resources

GPT go Subscribers:

- Exclusive experiences: Access to all premium models, and developer tools to provide an elevated experience.

Pay-as-You-Go Users:

- Value-based pricing and tiered options
- Discounts on multiple purchases or loyalty discounts

Freemium Users:

- Non-intrusive ads
- Limited free trials and introductory offers

TARGETING STRATEGIES

VALUE PROPOSITION

PremiumGPT experiences

Developer opportunities

User-centric Pricings

MESSAGING AND POSITIONING

DIFFERENTIATION

Unique combination of Subscription Models

Tiered pricing

Quality driven GPT Store

GPT STORE OPTIMIZATION

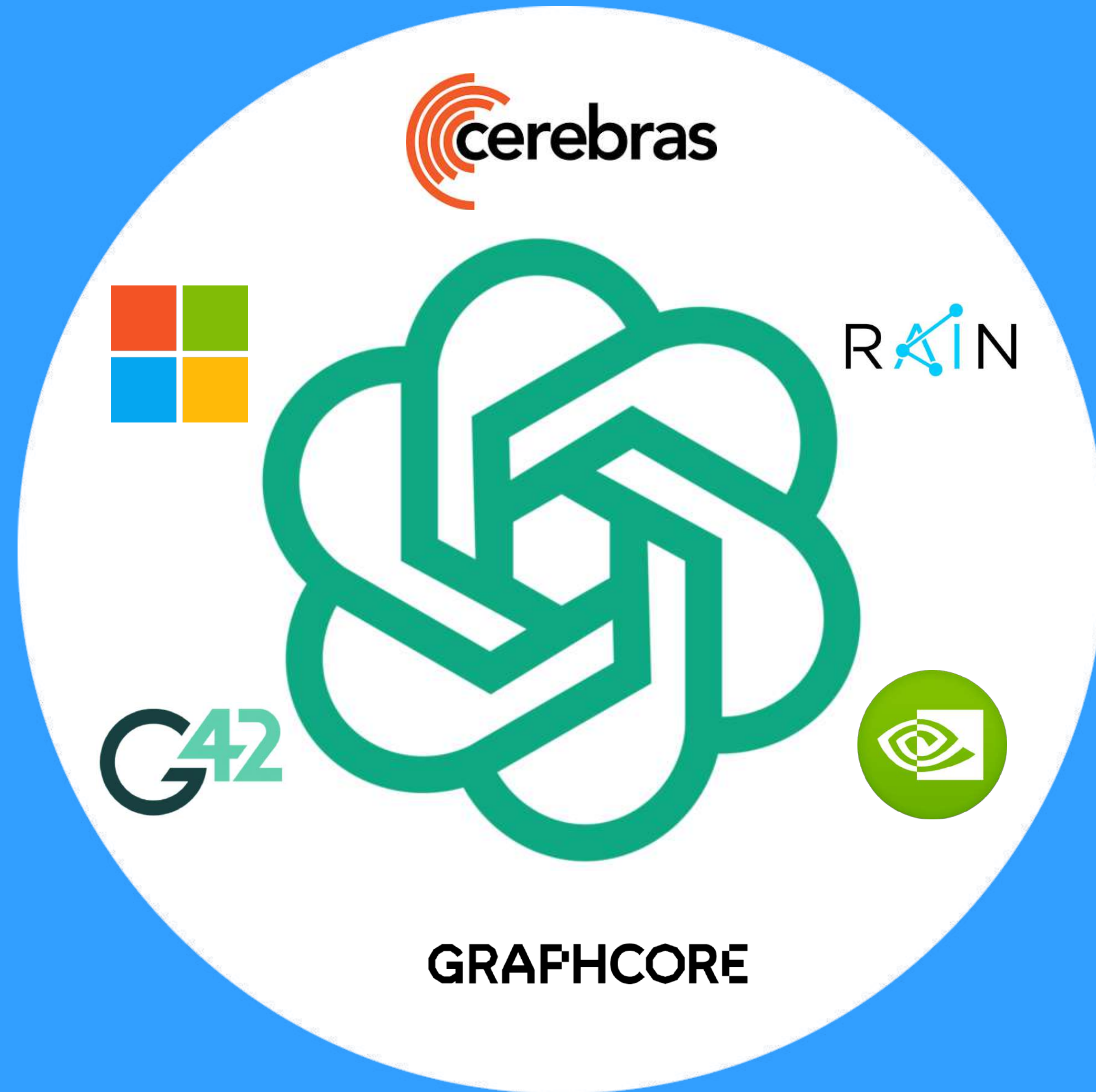
Easy Navigation to Discover categorized GPTs

Comprehensive Reviews and Ratings for decision making

Personalized recommendations based on user preference



Revenue Models	Financial Feasibilty	Over Saturation control & Quality ensurance	Success Metrics	Efforts	Impact	Priority
1. Subscription from user	- Freemium Features for users - Plans starting from \$25, assuming a user have 500 monthly prompts = $0.04\\$ \times 500 = \\20 (operating cost), remaining 5\$ is shared between Open AI and GPT owner (Developer)	- Set number of free trials following GPT Builder's guide. - Limited features access during free trials. - Read comments and check ratings before buying any GPT. - Feedback system to help the developer improve models.	- Number of subscriptions for each tier. - User ratings and feedback. - Number of trials taken by users.	Medium	High	#1
2. One time Payment from Developer	- Different pricing tiers based on GPT performance. - Payment of minimal one-time fees to publicly monetize GPTs. - Student discount available to foster innovation	- Registration fee for developers to prevent spams and ensure quality. - Student Discount to cultivate inventive thinking. - Payment of one-time developer registration fee to keep quality in check.	- Number of developers making one-time payments. - Positive impact on GPT quality and innovation. - Student participation and feedback.	Medium	High	#1
3. Data Collection and Analysis	- Monetization of anonymized user interaction data - Varied revenue generation according to Data Quality - One time financial effort in setup and planning, then enjoy the revenue	- Leveraging anonymized user data to prioritize privacy. - Gathering data to drive platform improvement. - Use statistical insights for monetization, safeguarding personal data.	- Revenue generated from data analytics and challenges. - Improvement in GPT models through data-driven enhancements.	High	High	#3
4. Advertisements	- Advertisement implemented only if necessary for revenue generation - Good revenue source, but initial strategy investments are high	- Clean UI/UX prioritized; advertisements considered only if necessary. - Student discount to balance user experience. - Ads integrated strategically within model outputs. - Relevant advertisements in GPT Store categories.	- Advertisement revenue. - User satisfaction and engagement. - Student participation and feedback.	High	Medium	#4
5. GPT Gauntlet	- Registration fee for GPT Gauntlet participation. - Revenue generated from the GPT Store Enterprise plan.	- Tiered GPTs based on performance to prevent oversaturation. - User rating system for GPTs to manage performance. - GPTs featured in GPT Gauntlet based on user ratings. - Anonymized user data used for research and development.	- Revenue from GPT Gauntlet participation. - User satisfaction and engagement. - Success of GPTs in the Gauntlet.	Low	Medium	#2



OPEN AI's AI CHIP DESIGN FIRM / STARTUP ONBOARDING

PROBLEM STATEMENT

- Challenge:** Entering the capital-intensive **AI chip market** solo is high-risk for OpenAI.
- Opportunity:** Onboarding a partner offers access to resources, market reach, and expertise
- Key Metrics:** Economics and market synergies, aligned goals, and cultural fit are crucial for success
- Feasibility:** Partnering **increases success odds** while balancing risk and control



Current AI Chip Industry

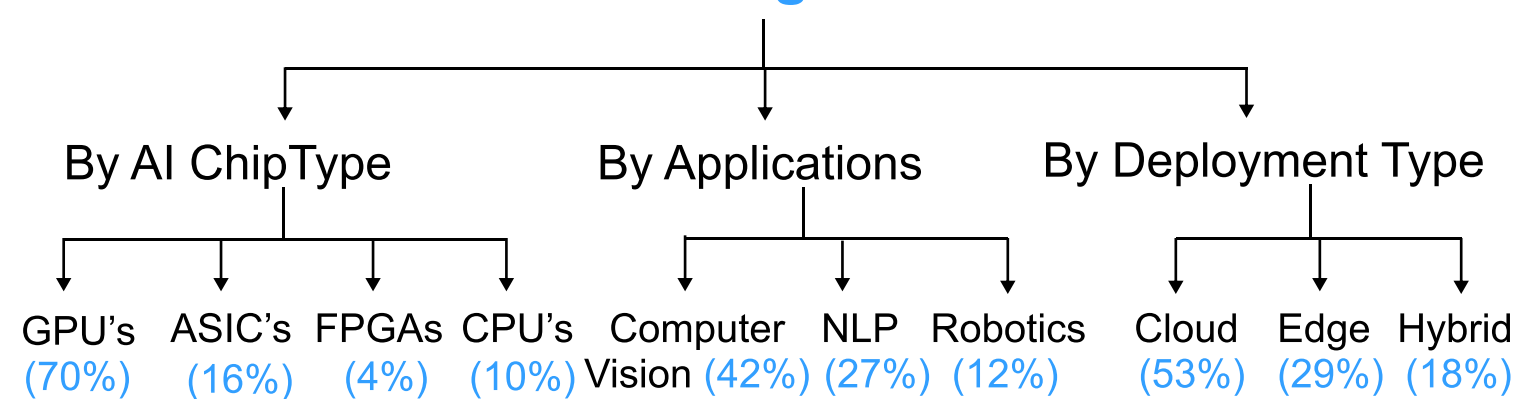
Current AI Chip Industry
in 2023

\$ 21.87 Billion

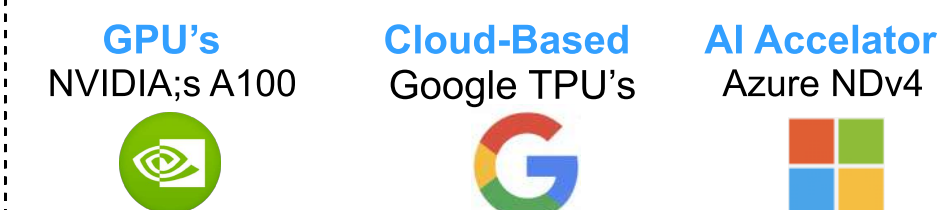
Projected AI Chip Industry
in 2030 (29.2% CAGR)

\$ 135.18 Billion

Market Segmentation



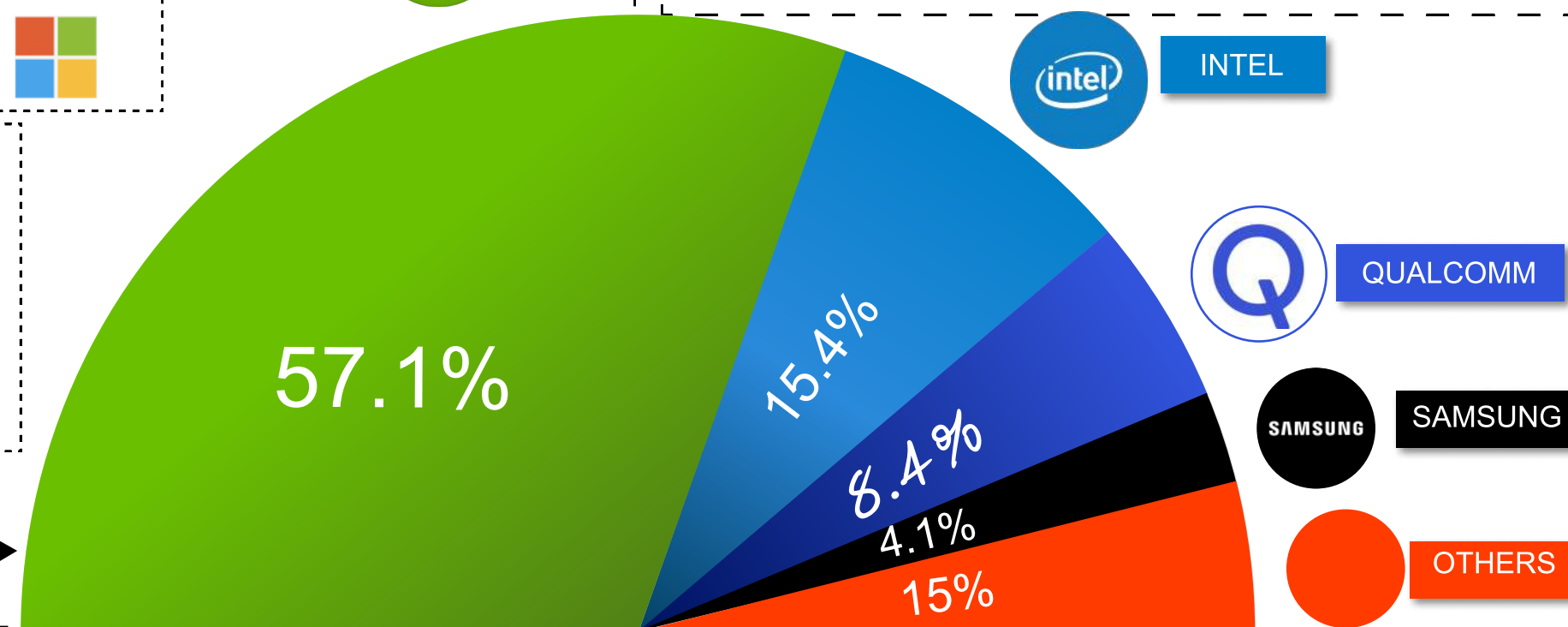
Open AI's current usage



Emerging Players



Market Share



What is an AI Chip?

An AI chip, also known as an AI accelerator or AI hardware, is a specialized piece of computer hardware designed to efficiently handle the demanding computational needs of artificial intelligence (AI) tasks. Unlike traditional CPUs that excel at diverse tasks, AI chips are optimized for specific types of computations commonly used in AI algorithms

Number of AI Chip Startups: Over **500 AI chip** startups have emerged globally, representing a diverse and dynamic ecosystem

Factors fueling AI Chip Industry

Booming AI Adoption

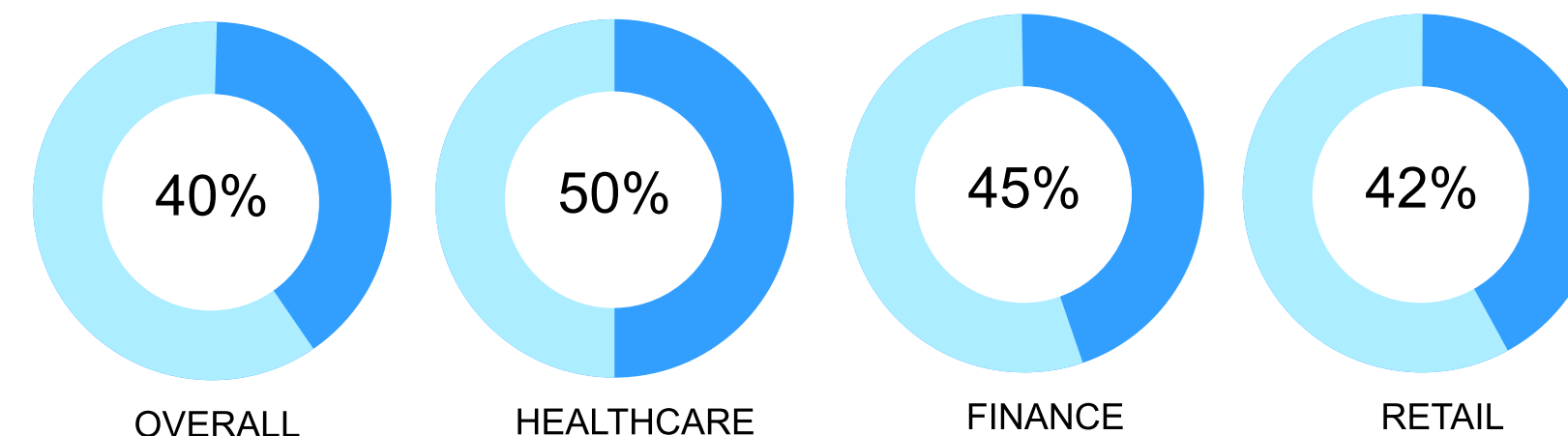
Diversification of AI Application

Performance limitations of CPUs

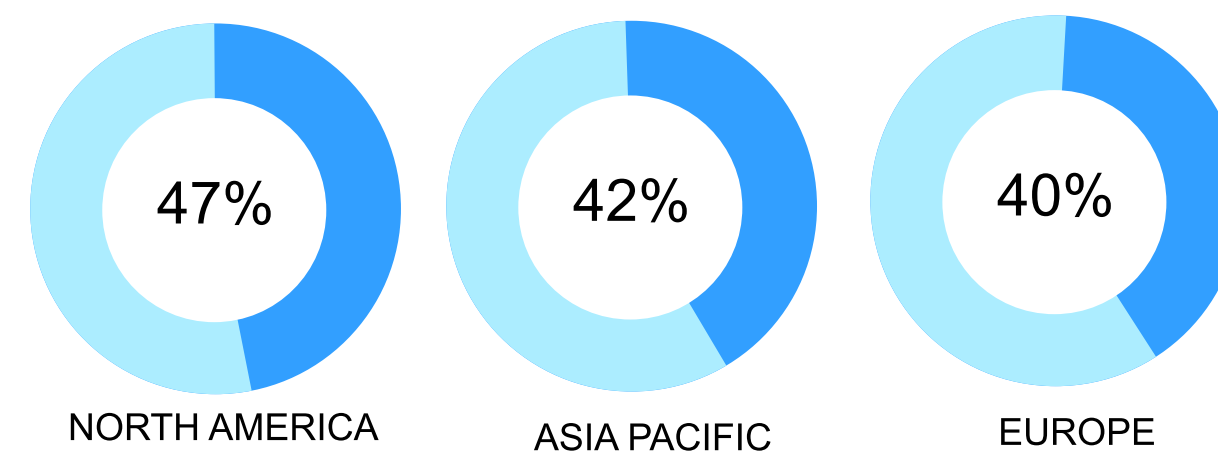
Market Opportunity and Key Metrics

AI Adoption Rate

Industry-Specific Adoption:



Regional Variation:



Average Revenue per Unit (ARPU): \$1,000 per chip by 2030

- High-performance GPUs for AI training might reach \$5,000 per chip
- Edge AI chips for low-power applications could be around \$100 each

VC investment in AI chip startups is booming, reaching over **\$15 billion globally in 2023**

Global AI chip development Investments are projected to reach **\$100 billion by 2025**

Emerging Trends:

- Edge AI:** Growing demand for decentralized AI processing at the edge is driving the development of low-power, low-cost AI chips for devices like smart sensors and industrial robots.
- Neuromorphic Computing:** This technology, mimicking the human brain, holds promise for more efficient and natural AI processing in the future.
- Open-Source Hardware:** Collaboration and shared development through open-source initiatives are gaining traction, accelerating innovation and reducing costs.

LIMITATIONS

- High Development Costs:** Designing and manufacturing custom AI chips requires significant financial resources, creating a barrier for OpenAI, a non-profit research lab.
Estimated cost of making an AI Chip: \$100 M to \$1 B, depending on complexity and targets
- Time to Market:** Developing a new AI chip can take **2-3 years**, putting OpenAI behind established players with existing technology. Delay research progress and limit early impact.
- Risks of Technology Failure:** Estimated financial loss could range from **\$10 M to \$100 M** depending on the severity and stage of development.
- Performance Benchmark:** OpenAI's chip needs to achieve at least 15% higher performance or 20% higher efficiency compared to existing options to justify its development and adoption.

OVERVIEW





ChatGPT-owner OpenAI is exploring making its own AI chips
06 Oct 2023, Anna Tong, Reuters
CEO Sam Altman has made the acquisition of more AI chips a **top priority** for **Open AI**.

- Why?*
- 1. A shortage of the advanced processors that power OpenAI's software
 - 2. **“Eye-Watering”** costs associated with running the necessary hardware
 - 3. Scarcity of GPUs, a market dominated by **Nvidia**, which controls **80%** of the global AI Chip market

CURRENT SCENARIO

- Since past few years **Open AI** has discussed various options to solve the shortage of **expensive AI chips** that OpenAI relies on but **Open AI** has not yet decided to move ahead to build its own AI chip
- Since 2020, **OpenAI** has developed its generative artificial intelligence technologies on a massive supercomputer constructed by **Microsoft**, one of its largest backers, that uses 10,000 of **Nvidia's** GPUs.
- OpenAI's main backer, **Microsoft**, is also developing a **custom AI chip** that **OpenAI** is testing,
- Demand for specialized AI chips has soared since the launch of **ChatGPT** last year. **Specific chips, or AI accelerators**, are necessary to train and run the latest generative AI technology. **Nvidia** is one of the few chipmakers that produces useful AI chips and dominates the market.
- An acquisition of a chip company could speed the process of building OpenAI's own chip as it did for **Amazon.com** and its acquisition of **Annapurna Labs** in 2015.

POTENTIAL RISKS

Building it's own chip could amount to hundreds of millions of dollars a year in costs. Even if OpenAI committed resources to the task it would not guarantee success.

Even if OpenAI goes ahead for a custom chip including an acquisition the effort is likely to **take several years** leaving the company dependent on commercial providers like **Nvidia** and **Advanced Micro Devices(AMD)**

CONCLUSION: **OpenAI's ChatGPT** fuses a **Microsoft** supercomputer that uses **10,000 Nvidia (GPUs)**. While **Nvidia** is necessary, the dependency may need to be revised in the long term because, each **ChatGPT** query costs around 4 cents. Even if OpenAI plans include a **custom chip** – by acquisition of Startups like **Rain/Cerebras** or with the help of **Microsoft** or **other strategies** – the effort will likely take several years. In short, **OpenAI** would still be **highly dependent** on **Nvidia** for a while and have to bear around **\$7,00,000 daily running cost** of **Chat GPT**.



OpenAI agreed to buy \$51 Million of AI Chips from a Startup “RAIN AI” backed by CEO Sam Altman
13 Dec 2023, Paresh Dave, WIRED

Rain is based less than a mile from **OpenAI's** headquarters in San Francisco and is working on a chip it calls a **neuromorphic processing unit (NPU)**, designed to replicate features of the human brain.

CURRENT SCENARIO

Rain told investors that Altman had personally invested more than \$1 million in the company.

Rain could get its first hardware to customers as early as October next year.

“**Rain** is poised to build a product that will define new **AI chip** markets and massively disrupt existing ones.” Sam Altman wrote, “Moving forward I will continue to help Rain in every way I can.”

- FACTORS SUPPORTING INVESTMENT IN RAIN AI**
- In a **Rain** press release for the fundraise last year, Altman applauded the startup for taping out a prototype in 2021 and said it **“could vastly reduce the costs of creating powerful AI models** and will hopefully one day help to enable true artificial general intelligence.”
 - **Rain**, founded in 2017, has claimed that its brain-inspired **NPU**s will yield potentially **100 times more computing power** and, for training, **10,000 times greater energy efficiency than GPUs**, the graphics chips that are the workhorses for AI developers such as **OpenAI** and primarily sourced from **Nvidia**.
 - **Rain's** initial chips are based on the RISC-V open source architecture **endorsed by Google, Qualcomm**, and other tech companies and aimed at what the tech industry calls edge devices, located far from data centers, such as phones, drones, cars, and robots. Rain aims to provide a chip capable of both **training machine algorithms** and running them once they're ready for **deployment**.

CONCLUSION: **RAIN AI** is a potential acquisition target



Microsoft unveils custom AI chip, with help from OpenAI, playing catch-up with Amazon
15 Nov 2023, Todd Bishop, GeekWire



Cerebras Systems introduced the CS-2 AI system, featuring the record-breaking 2nd-gen Wafer Scale Engine with 850,000 cores and 2.6 trillion transistors. The company, backed by \$720 million in funding from VC firms and notable investors like Sam Altman (OpenAI CEO) and Andy Bechtolsheim (Sun Microsystems Founder), deploys 64 of its \$1.5 million chips at a site.



Published on October 16, 2023 In AI Origins & Evolution
Why OpenAI should acquire Graphcore
OpenAI wants to make its own AI chips and Graphcore wants to make some money. What else does the partnership need?



What OpenAI's latest batch of chips says about the future of AI
OpenAI has secured more H100 chips and is expecting even more soon
Michelle Cheng, November 14, 2023
What do more chips mean for OpenAI?
Altman did not disclose a timeline for the release of **GPT-5** but the quick succession of release, with GPT-4 coming just eight months ago, following the release of its predecessor GPT-3 in 2020, highlights a rapid development cycle.

Nvidia's H100 chips, which are tailored for **generative AI** and run much faster than previous chip models. The more powerful the chips, the faster you can process queries.

One could say that the level of attention on AI chatbots like **OpenAI's ChatGPT** and **Google's Bard** this year matches the amount of focus on **Nvidia's \$40,000 H100 chips**. OpenAI, like many other AI companies, uses Nvidia's latest model of chips to train its models.

The procurement of more chips from OpenAI signals that more sophisticated AI models, which go beyond powering the current version of chatbots, will be ready in the near future.



Potential AI Chip Organization Comparision							
Chip	About	Performance (FLOPS, Inference Speed, Memory Bandwidth, Tensor Cores)	Efficiency (Power Consumption, Price-Performance)	Architecture (ISA, Interconnect, Programming Environment)	Availability (Production Volume, Lead Times, Market)	Support and Maintenance	OpenAI Fit Analysis
NVIDIA H100	NVIDIA's flagship AI accelerator for high-performance computing and cloud data centers.	53.7 PetaFLOPS, 312 ms/image (ResNet-50), 760 GB/s, 7th Gen Tensor Cores	Moderate (350W), \$5/TFLOPS	CUDA, NVLink, C++, Python	Widely available, short lead times	Strong	Good, strong tensor cores, CUDA compatibility
Microsoft Azure Machine Learning (custom chip)	Microsoft's custom-designed AI chip for Azure cloud workloads, details are limited.	Unknown	Unknown	ARM, Unknown, C++, Python	Limited, not publicly available	Strong	Potential, tailored for Azure ML workloads, limited information
Rain AI	Startup developing a highly efficient AI chip for edge and cloud applications, with a focus on open-source software and hardware.	Up to 1 ExaFLOPS (estimated), N/A, N/A, Custom	High (efficiency focus)	ARM, Custom, C++, Python	Limited, pre-release, niche market	Emerging	Potential, innovative architecture, open-source focus
Cerebras CS-1	Cerebras' wafer-scale AI chip, designed for extreme performance in large-scale AI workloads.	415 PetaFLOPS, N/A, 22 TB/s, Custom	Moderate (7.5 kW)	Custom ISA, Wafer Scale Engine, Python	Limited, long lead times, niche market	Limited	Niche, extreme performance, high cost
Graphcore IPU	Graphcore's AI accelerator optimized for graph-based AI models, offering high performance and efficiency.	Up to 50 PetaFLOPS (estimated), N/A, 600 GB/s, Custom	High (efficiency focus)	C++, Python	Limited, pre-release, niche market	Emerging	Potential, innovative architecture, good for graph-based models
Abu Dhabi G42	A high-performance AI chip developed in Abu Dhabi, aiming for exascale performance.	Up to 1.5 ExaFLOPS (estimated), N/A, 2.4 TB/s, Custom	Moderate	ARM, Custom, C++, Python	Limited, pre-release, niche market	Emerging	Potential, high performance, limited information
Tigris	A low-power AI chip designed by Meta for potential use in AR/VR devices, details are scarce.	N/A, N/A, N/A, N/A	N/A, N/A	ARM, Unknown, C++	Limited, primarily used in specific devices	Limited	Uncertain, limited information and application, potentially low power for mobile devices

OPENAI'S AI CHIP ONBOARDING STRATEGY

Target early stage startups with innovative architectures, expertise, and strong teams

Consider various partnership models:

- 1. **Acquisition:** Full control, high upfront investment.
- 2. **Joint Venture:** Shared risks and rewards, access to expertise and infrastructure.
- 3. **Strategic Investment:** Funding for access to tech, talent, and collaboration.

MUTUAL BENEFITS

- **Benefits for OpenAI:**
 - Reduced cost and risk (shared development).
 - Faster time to market (leveraging existing expertise).
 - Enhanced technology (combining AI know-how and chip design expertise).
 - Talent acquisition (attracting top talent from the startup ecosystem).
- **Benefits for the Startup:**
 - Financial resources (access to OpenAI's funding and resources).
 - Credibility and market access (leveraging OpenAI's reputation and user base).
 - Technical collaboration (optimizing chip design for real-world AI workloads).

FINAL STRATEGY

ROADMAP TO ONBOARDING

Maintain and strengthen the ongoing partnership with NVIDIA for chip supply continue optimizing GPT models for compatibility with NVIDIA chips.

For time being prime focus to be on the success and growth of the GPT Store to generate substantial revenue for investment in the chip industry

Along with focus on growth of GPT Store, OpenAI should simultaneously fund the R&D division of the blooming startups like **Cerebras, G42, Graphcore, Rain**, etc from the investments it receives from tech giants like Microsoft to fund talent and innovation in this domain

Once OpenAI is ready for the final onboarding the next step involves evaluating all the potential startups and figuring out which has the best matching synergics with OpenAI through thorough market research

Next step involves integrating newly acquired AI chips into the development pipeline by conducting extensive testing to ensure compatibility and optimal performance with OpenAI's AI models and hence gradually transitioning from the reliance on NVIDIA chips to the new advanced AI chips

FEASIBILITY AND EXPANSION

- **Maturing Startup Landscape:** Promising startups ideal for acquisition/partnership.
- **Open-Source Culture:** Alignment with OpenAI's values facilitates collaboration.

Examples :

- **RAIN AI:** Potential partner for accessible and efficient AI chips (open-source focus).
- **Graphcore:** Expertise in graph-based AI models (applications in NLP or scientific computing).
- **Cerebras:** Collaboration on specific architectural aspects (extreme performance chips).

ECONOMIC SYNERGIES

- Reduced Costs & Shared Risks:** Sharing development burdens for financial feasibility.
- Resource Leverage:** Combining OpenAI's resources with startup expertise for efficiency.
- Market Expansion:** Diversifying customer base through established reach or niche markets.

MARKET SYNERGIES

- Target Market Alignment:** Immediate value through relevant niches and partnerships.
- Shared Vision & Goals:** Strong foundation for collaboration in responsible & accessible AI.

TECHNOLOGY SYNERGIES

- Efficient Architecture:** Choosing startups with chips aligning with OpenAI's standards.
- Integrated Expertise:** Partnership with teams blending AI skills and chip designing.
- Flexible Scalability:** Collaborating on adaptable chips for diverse AI applications.

FINAL CONCLUSION



EXECUTIVE SUMMARY

OpenAI's GPT Store Novel Revenue Model

ChatGPT has more than 99% free user base, resulting to around \$7,00,000 daily running cost, even though ChatGPT has other revenue streams they are hardly able to recover the cost.

The launch of GPT Store in January'24 and its strategic Monetization can result into more Revenue streams as mentioned

1. Subscription from User:

Plus Subscription: Basic subscription plan (Financially Feasible) as ChatGPT is currently offering and then paid tiered individual custom GPTs to ensure user flexibility.

GPTgo Subscription: Unlimited custom GPTs access like Google's "Play Pass"

2. One Time Payment from Developer:

One time fee barrier to ensure quality of custom GPTs and limited free trials allowed to test the ability of GPT Builder that can be used privately only.

3. Data Analytics and its Monetization:

Another possible stream of revenue, Open AI must take care of personal user data with monetizing the user behaviour data and sharing the type of data without any legal issue

4. Advertisements:

As ChatGPT has more than 99% free user base, it may be crucial to generate revenue from this large group of users at a cost of compromising with User's Experience

5. GPT Gauntlet (GPT building Competition)

Focusing on "GPT innovation" and less on profit, a low effort with a good impact model

Generative AI will be a booming Market in future, hence GPT Store has a huge potential to generate high Revenue and increase profits of Open AI for further developement

OPEN AI's AI CHIP DESIGN FIRM / STARTUP ONBOARDING

OpenAI is exploring ways to handle GPU shortage and the challenges of the capital-intensive and rapidly evolving AI chip market. The primary objectives are to reduce costs, mitigate risks, and accelerate time to market by leveraging external expertise and resources

Important features from our research:

- OpenAI targets early-stage startups for AI chip challenges.
- Diverse partnerships prioritizing architecture, market, and talent.
- Booming **\$778.7 billion** AI chip market leveraged by OpenAI.
- Startups like **RAIN AI, Graphcore, and Cerebras** identified for collaboration.
- Collaborative benefits: reduced costs, shared risks, resource leverage.
- Challenges: high development costs, technology risks, Nvidia dependence, requiring careful considerations.

Hence, we concluded :

It is best for OpenAI to invest on **R&D** of various AI chip startups like **Rain, Tigris & Abu Dhabi based G42**, which it is doing currently to resolve its GPU challenges in long term and for now it should continue its partnership with **Nvidia** till GPT store generates enough revenue to onboard the blooming startups.



References

1. <https://cio.economictimes.indiatimes.com/news/investments/openai-in-talks-to-raise-new-funding-at-100-billion-valuation-report/106227382>
2. <https://www.wired.com/story/openai-buy-ai-chips-startup-sam-altman/>
3. <https://cio.economictimes.indiatimes.com/news/investments/openai-in-talks-to-raise-new-funding-at-100-billion-valuation-report/106227382>
4. <https://fortune.com/2023/08/30/chatgpt-creator-openai-earnings-80-million-a-month-1-billion-annual-revenue-540-million-loss-sam-altman/>
5. <https://www.indiatoday.in/technology/news/story/openai-could-go-bankrupt-by-2024-as-chatgpt-costs-over-rs-5-crore-to-operate-everyday-2420302-2023-08-13>
6. <https://www.statista.com/statistics/1365145/artificial-intelligence-market-size/#:~:text=Global%20artificial%20intelligence%20market%20size%202021%2D2030&text=According%20to%20N>
7. <https://www.mckinsey.com/capabilities/quantumblack/our-insights/generative-ai-is-here-how-tools-like-chatgpt-could-change-your-business>
8. <https://www.reuters.com/technology/openai-delays-custom-gpt-stores-launch-axios-2023-12-01/>
9. <https://www.bloomberg.com/news/articles/2023-08-28/openai-unveils-chatgpt-for-business-customers-stepping-up-revenue-push>
10. <https://bestmediainfo.com/2023/01/microsoft-invests-10-billion-in-openai-to-accelerate-ai-breakthroughs-and-deploy-them-on-azure>



You

"THANK YOU"



ChatGPT

You're very welcome! If you have any further questions or if there's anything else I can assist you with, feel free to let me know.



TEAM CASE ACE



SHRITIK JAISWAL



RUGVED BELE



MANISHA SAHOO



AKSHARA SHUKLA



NAVEEN RAIYA